

Hexadecimal Addition

$$\text{base}(r) = 16$$

$$0 \text{ to } (16-1)$$

$$0 \text{ to } 15$$

$$26 = \underline{1} \times 16 + \underline{10}$$

$$r = 26 - 16$$

$$r = 10$$

$$1 + 13 + 10 = 24$$

$$13 \times 16^0 + 13 \times 16^0$$

$$26 \times 16^0$$

$$(16 + 10) \times 16^0$$

$$16 \times 16^0 + 10 \times 16^0$$

$$1 \times 16^1 + 10 \times 16^0$$

carry

0
1
2
3
4
5
6
7
8
9
A
B
C
D
E
F

10
11
12
13
14
15

i)

$$\begin{array}{r} 5 \ 6 \ 8 \ 9 \\ + 4 \ 5 \ 7 \ 4 \\ \hline 9 \ B \ F \ D \end{array}$$

Ans

$$\begin{array}{r} F \\ F \\ + A \\ \hline 40 = 2 \times 16 + 8 \end{array}$$

ii)

$$\begin{array}{r} 16^2 \quad 16^1 \quad 16^0 \\ \downarrow \quad \downarrow \quad \downarrow \\ 1 \quad 1 \\ + \quad A \quad D \quad D \\ \hline 1 \quad 8 \quad 8 \quad A \end{array}$$

Ans

$$24 = 1 \times 16 + 8$$

$$\begin{array}{r} F \\ + F \\ \hline 30 = 1 \times 16 + 14 \end{array}$$

$$2 \times 16 = 32$$

4	
5	
6	
7	
8	
9	
A	10
B	11
C	12
D	13
E	14
F	15

$$\begin{array}{r}
 5 \ 6 \ 8 \ 9 \\
 + \ 4 \ 5 \ 7 \ 4 \\
 \hline
 9 \ B \ F \ D \\
 \hline
 \text{Ans}
 \end{array}$$

$$\begin{array}{r}
 A \ D \ D \\
 + \ D \ A \ D \\
 \hline
 1 \ 8 \ 8 \ A \\
 \hline
 \text{Ans}
 \end{array}$$

$$\begin{array}{r}
 F \) \ 30 \\
 F \) \ 10 \\
 + \ A \\
 \hline
 40 = 2 \times 16 + 8
 \end{array}$$

$$\begin{array}{r}
 F \\
 + \ F \\
 \hline
 30 = 1 \times 16 + 14 \text{ E}
 \end{array}$$

$24 = 1 \times 16 + 8$
 $2 \times 16 = 32$

$x = 26 - 16$
 $x = 10$
 $L + 13 + 10 = 24$
 $13 \times 16^0 + 13 \times 16^0$
 26×16^0
 $(16 + 10) \times 16^0$
 $16 \times 16^0 + 10 \times 16^0$
 $1 \times 16^1 + 10 \times 16^0$
carry

H.w i)

$$\begin{array}{r}
 8 \ 9 \ 9 \\
 + \ 1 \ 8 \ 9 \\
 \hline
 \hline
 \hline
 \end{array}$$

A22

ii)

$$\begin{array}{r}
 D \ A \ F \\
 + \ B \ A \ F \\
 \hline
 \hline
 \hline
 \end{array}$$

195E

Hexadecimal Subtraction

Show

$$19 - 7 = 12 \quad B < C$$

$$22 - 8 = 14$$

$$16 + 6 \quad 16 + 3 \quad 16 + 11 = 27 - 12$$

ii)

$$\begin{array}{r} 896734B \\ - 587C \\ \hline 3ECF \end{array}$$

$$\begin{array}{l} 16 \times 16^6 \\ \hline 16 \times 16^1 \\ \hline 16 \times 16^0 \end{array}$$

Q:-3

$$\begin{array}{r} 7788 \\ - DEF \\ \hline \end{array}$$

6999

i)

$$\begin{array}{r} 9654 \\ - 5321 \\ \hline 4333 \end{array}$$

Ans

Q:-2

$$\begin{array}{r} FAB \\ - AEF \\ \hline \end{array}$$

4BC

Q:-1

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
A	- 10
B	- 11
C	- 12
D	- 13
E	- 14
F	- 15

Hexadecimal Multiplication

$18 = 1 \times 16 + 2$

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
A	- 10
B	- 11
C	- 12
D	- 13
E	- 14
F	- 15

i)

16^2

16^1

16^0

↓

↓

↓

1

9

4

x

1

2

1

2

8

9

4

x

A

6

8

Ans

$\underline{1} \times 16^1 + \underline{4} \times 16^0$
 $\underline{4} \times 16^1$

ii)

16^4

16^3

16^2

16^1

16^0

1

7

7

B

4

↑

↑

↑

↑

↑

16^4

16^3

16^2

16^1

16^0

Ans

24.30 i)

$\begin{array}{r} 818 \\ \times 21 \\ \hline \end{array}$

ii)

$\begin{array}{r} DAF \\ \times 54 \\ \hline \end{array}$

$36 = 2 \times 16 + \underline{4}$

$35 = \underline{2} \times 16 + \underline{3}$

$32 = \underline{2} \times 16 + 0$

$24 = \underline{1} \times 16 + 8$

$23 = \underline{1} \times 16 + \underline{7}$

$21 = 1 \times 16 + 5$

- 1. 10B18
- 2. 47D6C